Enhancing Human-Machine Collaboration in Industrial Workflows using Generative AI Models

Dissertation

(Abstract Viva)

in partial fulfilment of the requirements for the degree:

M.Tech. in Artificial Intelligence & Machine Learning

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Rounaq Choudhuri

I am driven by a passion for converting data into practical business intelligence.

My expertise includes a robust background in computer vision, where I have conceptualized and built video-based analytical solutions.

Currently, at LTIMindtree, I am focused on creating real-time intelligent surveillance systems for applications such as Environmental, Health, and Safety (EHS) and quality inspection.

My enthusiasm lies in crafting cutting-edge data solutions and utilizing the power of AI/ML and Generative AI to tackle intricate business problems.

Dissertation Overview

Overview

- This dissertation addresses the critical industrial challenge of limited anomaly and failure data, which hinders robust AI system operation and comprehensive human operator preparedness.
- The work proposes to leverage non-Transformer based Generative AI models, specifically Variational Autoencoders (VAEs) for robust anomaly detection from normal data, and Generative Adversarial Networks (GANs) for synthesizing diverse, realistic industrial data.
- The ultimate goal is to overcome data scarcity, enable more reliable AI systems, enhance human situational awareness and decision-making, and thus significantly improve human-machine collaboration for industrial efficiency and safety.

Motivation & Benefits

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- **What is the motivation?**
  - ✓ Industrial workflows are often very complex pertaining to the wide variety of intrinsic and extraneous factors.
  - ✓ Very difficult to model these complex distributions, therefore, AI systems often fail to provide robust solutions.
- **What is the benefit of this work?**
  - ✓ A comprehensive and comparative study in AI systems for industrial workflow.
  - ✓ It will create a premise for addressing the issues in modelling these complex data.
- **What are the expected outcome?**
  - ✓ Robust AI systems which can correctly assess unforeseen circumstances.
  - ✓ Harnessing the power of SOTA Generative AI Models to create realistic and diverse synthetic industrial data.
- **How it can eventually enhance human-machine collaboration?**
  - ✓ Meeting increasing demands to have closed loop systems in the era of Industry 4.0.
  - ✓ Foster trust among industrial machinery operators, business personnels and AI systems.

Scope of Work

In-Scope

- **Generative Models:** Implementation and experimentation with VAEs and GANs on industrial datasets.
- **Anomaly Detection:** Implementation of anomaly detection models and their evaluation.
- **Data Generation:** Use generative AI models to generate synthetic faulty data and its evaluation.
- **Model Generalization:** Anomaly detection model generalization using synthetically generated data.

Out-of-Scope

- **Transformer based Models:** Implementation and application of Transformer based architectures for human-machine interaction or synthetic data generation.
- **Advance HMIs:** Designing and constructing full-scale, highly interactive Human-Machine Interfaces or complete control systems based on model outputs.

Concepts At Work

Feature Scaling

Min-Max Scaling

- Min-max scaling adjusts the data to fit within the range [0,1].

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

- To reverse the transformation, the minimum and maximum values of the original data must be recorded.

$$x = x'(\max(x) - \min(x)) + \min(x)$$

Z-Score Scaling

- Z-score scaling is a transformation that standardizes the data by centering it around the mean (μ) and scaling it based on the standard deviation (σ).
- After applying this transformation, the mean of the scaled data becomes 0, and the standard deviation becomes 1.

$$x' = \frac{x - \mu}{\sigma}$$

- $sigma$
- To reverse the transformation, the mean and standard deviation of the original data must be recorded.

$$x = x'\sigma + \mu$$

Performance Metrices

F_β Score

- In binary classification tasks the F_β score is a family of metrics used to evaluate a set of predictions.
- **Prediction Types** – True Positives (TP), True Negatives (TN), False Positives (FP) & False Negatives (FN)
- **Recall** – quantifies the proportion of actual anomalies correctly identified by the classifier.
- **Precision** – measures how many elements classified as anomalies are true anomalies.
- **F_β score** – balances recall and precision.

$$Recall = \frac{TP}{TP + FN} \qquad Precision = \frac{TP}{TP + FP}$$

$$F_\beta = \frac{(1+\beta^2)(Precision \cdot Recall)}{\beta^2 \cdot Precision + Recall}$$

Reconstruction Metrics

- There are numerous metrics for calculating the difference between a tensor and its reconstruction.
- **Mean Absolute Error** – measures the average absolute difference between predicted and true values.
- **Mean Squared Error** – calculates the mean square difference between predicted and true values.
- **Mean Absolute Percentage Error** – quantifies the percentage difference between predicted and true values.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |xi - \hat{x}_i| \qquad MSE = \frac{1}{n}\sum_{i=1}^{n} (x_i - \hat{x}_i)^2$$

$$MAPE = 100\% \times \frac{1}{n}\sum_{i=1}^{n} \left|\frac{x_i - \hat{x}_i}{x_i}\right|$$

Performance Metrices (contd.)

ROC-AUC

- **Receiver Operating Characteristic (ROC) curve** – A graphical representation of the performance of a binary classification model, displaying the trade-off between the true positive rate (sensitivity) and the false positive rate (1-specificity) for different classification thresholds.

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

PR-AUC

- **Precision Recall (PR) curve** – A graphical representation of a binary classification model's performance, specifically illustrating the trade-off between precision and recall at different classification thresholds.

$$Recall = \frac{TP}{TP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

Learning Paradigms

Supervised Learning

- Supervised learning is a type of machine learning where the algorithm learns from labelled examples.
- Supervised models performs regression or classification tasks on new, unseen data.
- In the context of deep neural networks, it involves training a model using input-output pairs, where the inputs are the features or attributes of the data, and the outputs are the corresponding labels or target values.
- The goal is to teach a function f(x) that maps the input features to the correct output labels.

Unsupervised Learning

- Unsupervised learning is a type of machine learning where the algorithm learns patterns or structures in the input data without explicit labels or target values.
- Unlike supervised learning, it does not rely on explicit labels or target values. Instead, it focuses on finding patterns or organizing the data to capture meaningful information without specific guidance.
- In the context of deep neural networks, it is often used for clustering, dimensionality reduction, or generative modelling tasks.
- The goal is to discover inherent structures, relationships, or representations within the data.

Dataset

Dataset

Skoltech Anomaly Benchmark (SKAB) *SKAB is a benchmark designed for evaluating anomaly detection algorithms using time-series data. It contains datasets with collective and point anomalies, which are often rare events in real-world industrial processes. The dataset is multivariate, collected from sensors installed on a testbed.*

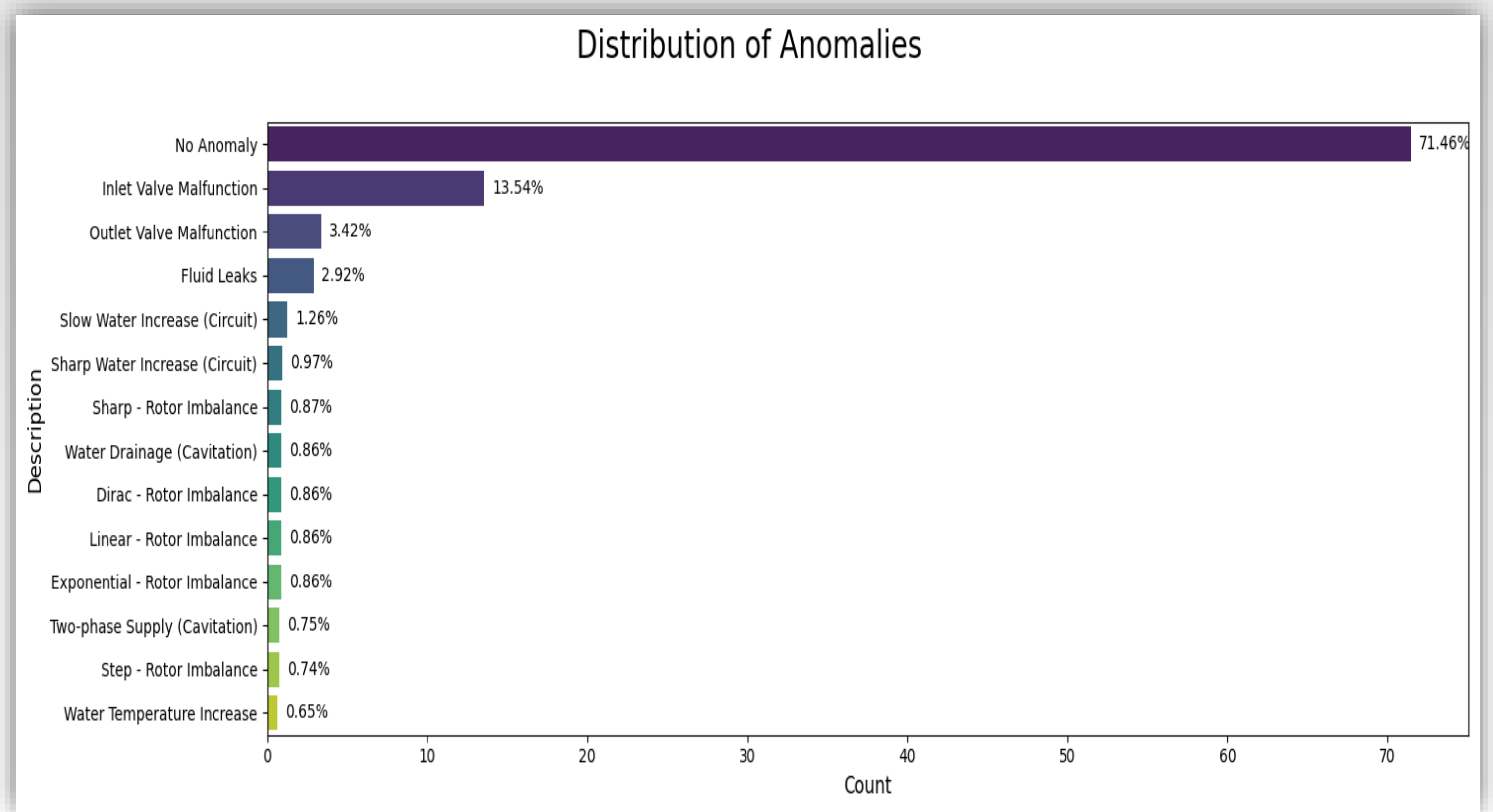
Download Link

Column Description

- **datetime :** Dates and times of data points (YYYY-MM-DD hh:mm:ss).
- **Accelerometer1RMS :** Vibration acceleration along the x-axis.
- **Accelerometer2RMS :** Vibration acceleration along the y-axis.
- **Current :** Current strength on electric motors (Amperes).
- **Pressure :** Pressure in the circuit after the water pump (Bar).
- **Temperature :** Temperature of the engine housing (◦C).
- **Thermocouple :** Temperature of the liquid in the circulation circuit (◦C).
- **Voltage :** Voltage on electric motors (Volts).
- **Volume Flow RateRMS :** The circulation flow rate of the fluid inside the loop (Liter per minute).
- **anomaly :** Shows whether the point is anomalous (0 or 1).
- **changepoint :** Shows whether the point is the point of change of state (0 or 1).

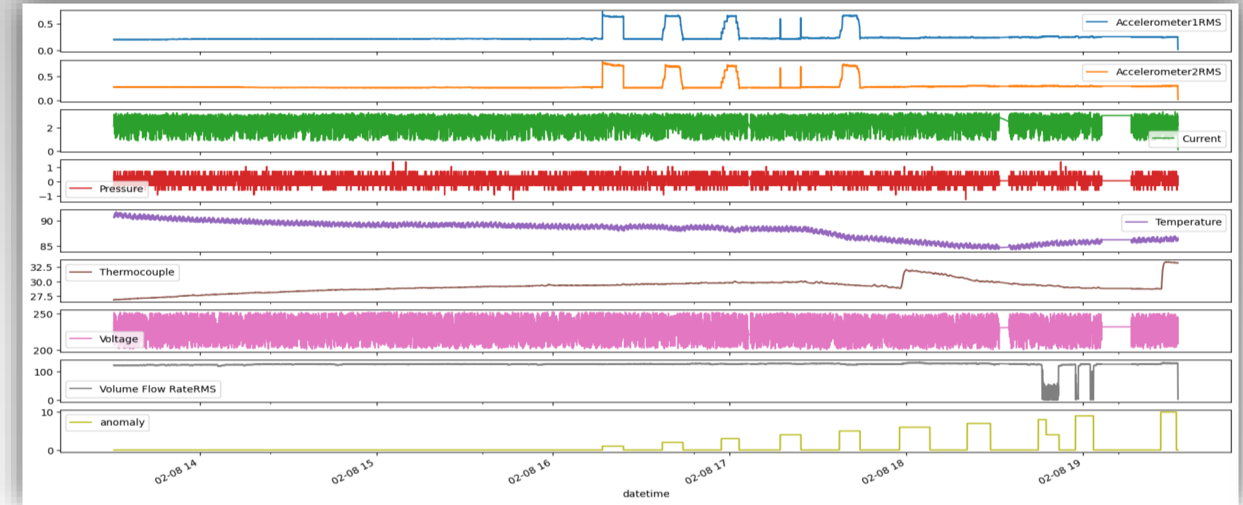
Data Proportion

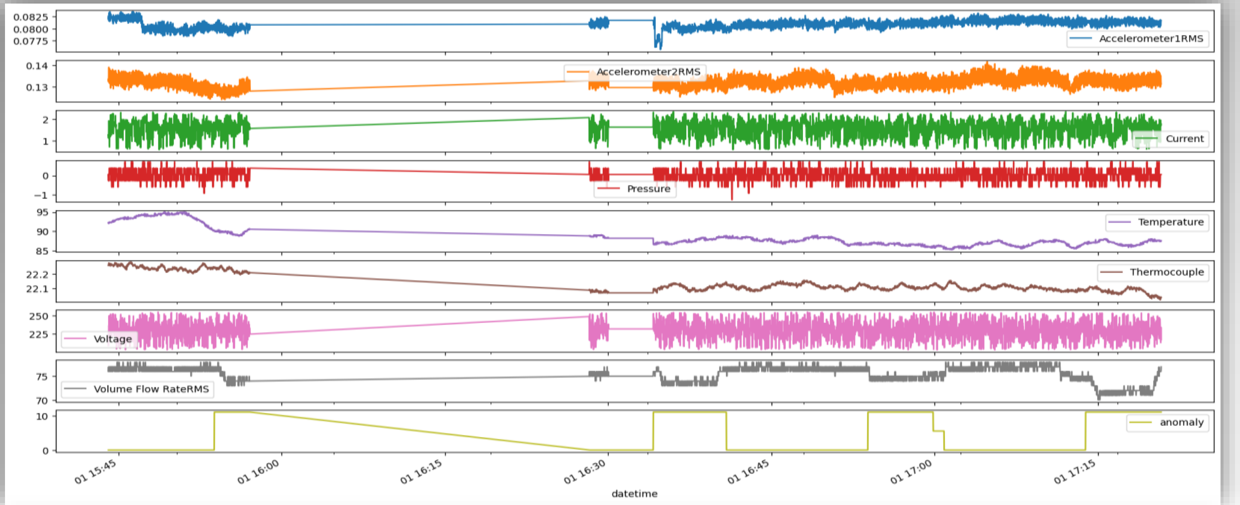
- A severe class imbalance within the dataset.
- “No Anomaly” instances constitute a dominant 71.46%.
- Anomalous data points are less than 30%.

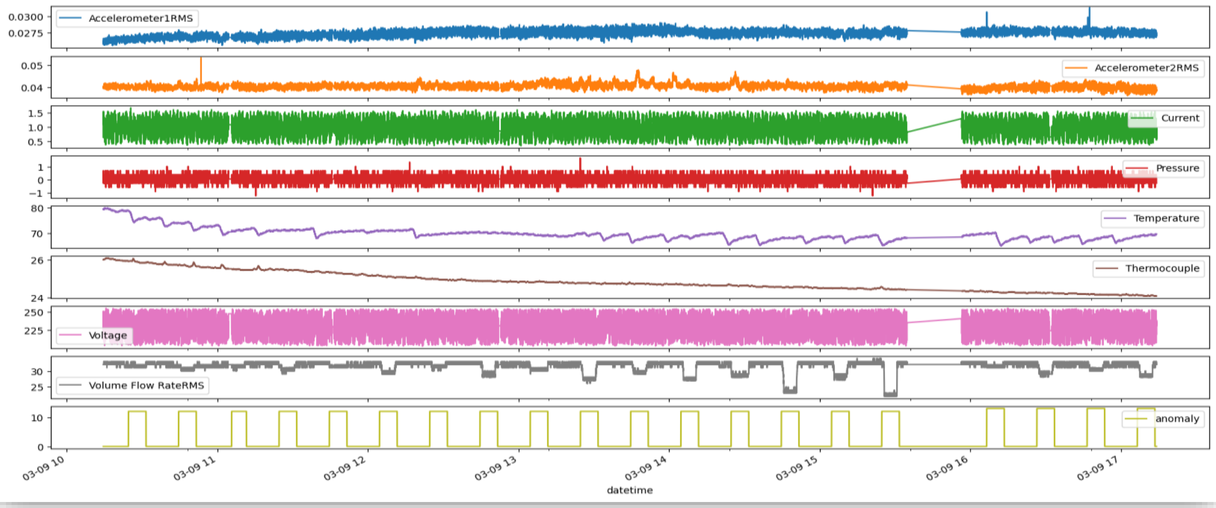


Data Preprocessing

Data Sampling & Merging







- Time series plots of the data set shows that the pump can operate at various operational states.
- Various operational status can host different types of anomalies.
- Not all features are shows significant deviation or spikes during anomalies.

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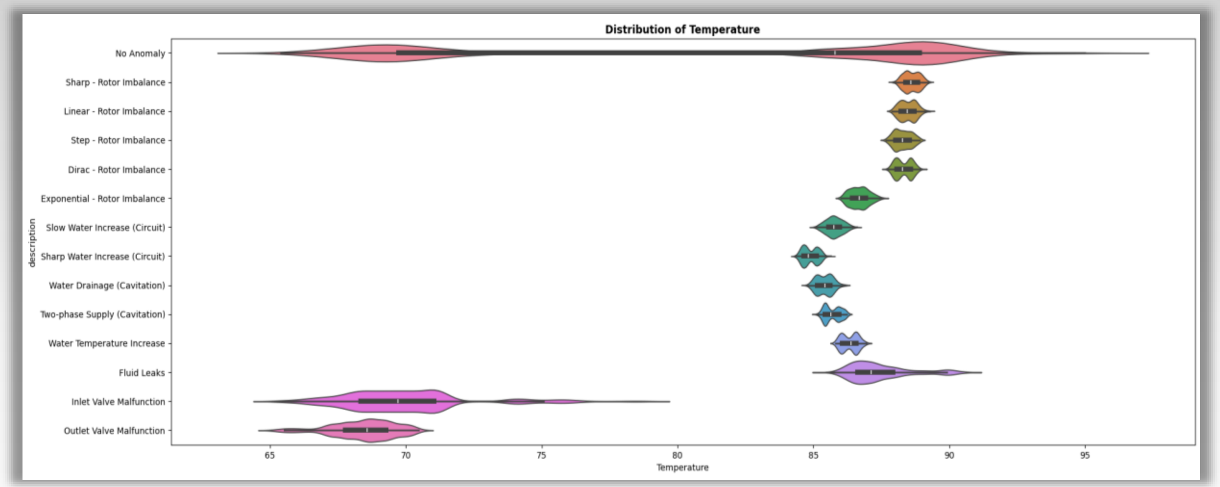
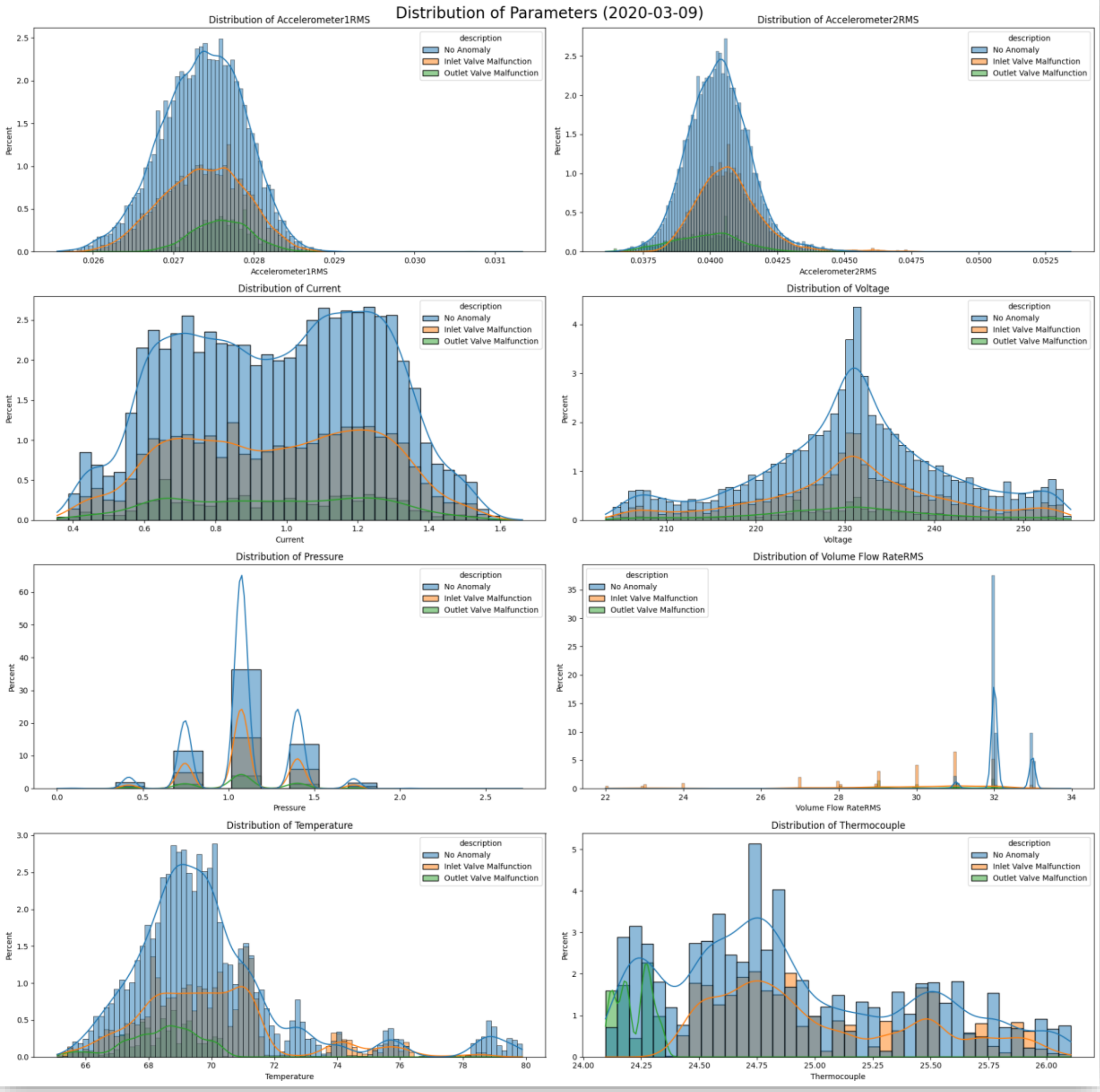
Exploratory Data Analysis

Univariate Analysis





- Histogram of most of the features during anomalies differ from each other.
- For some anomalies like Inlet & Outlet Valve Malfunction the histograms are overlapping with the normal data points.

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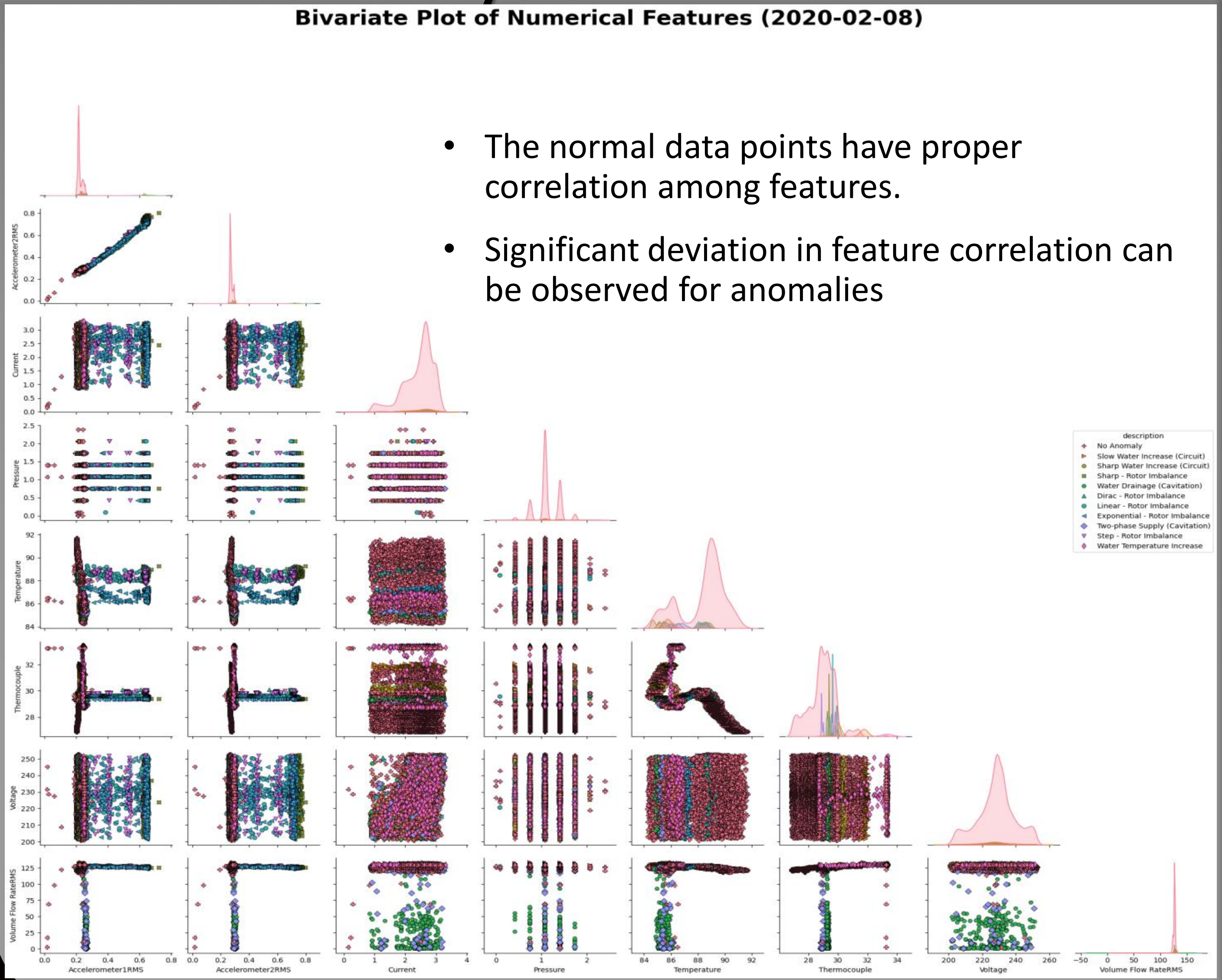
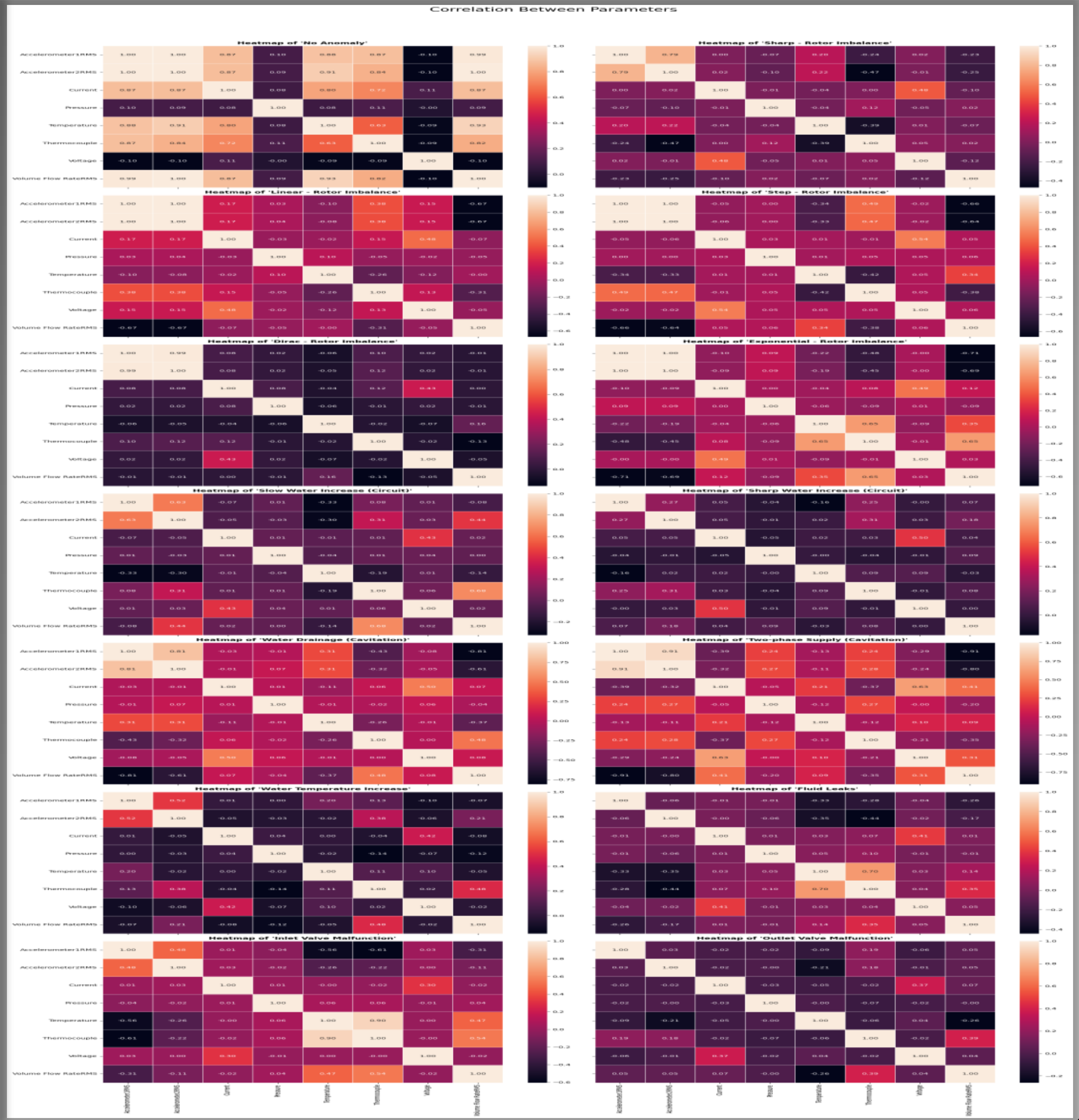
Bivariate Analysis



- The normal data points have proper correlation among features.
- Significant deviation in feature correlation can be observed for anomalies



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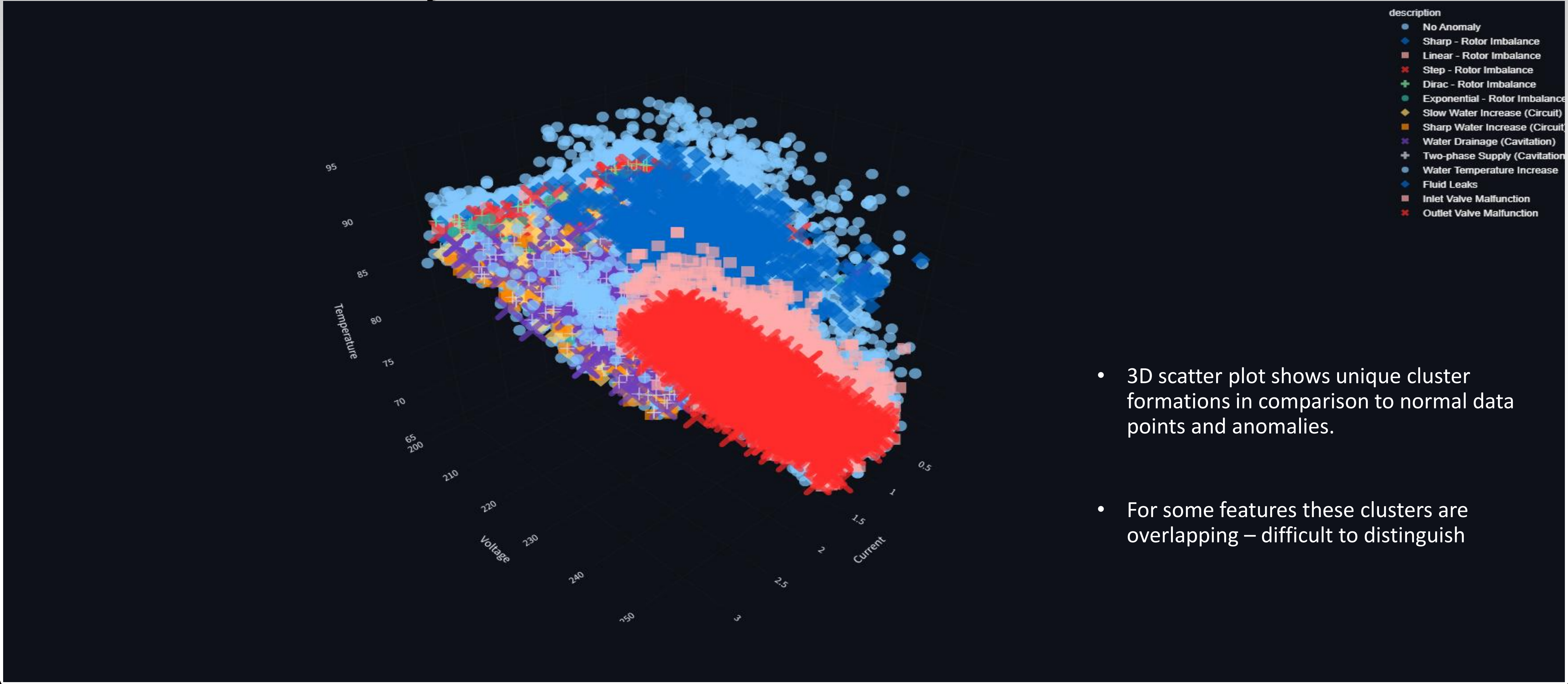
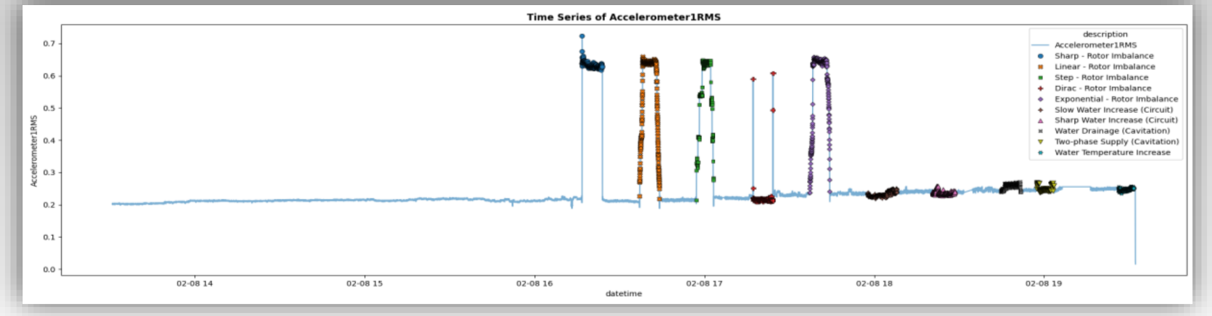
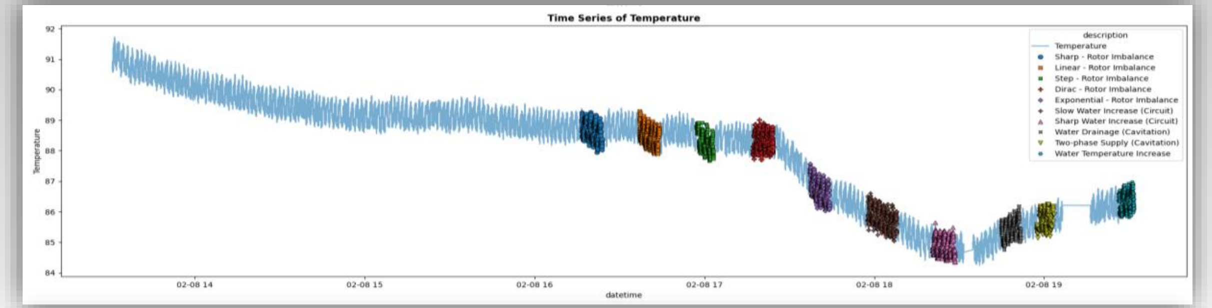
Multivariate Analysis

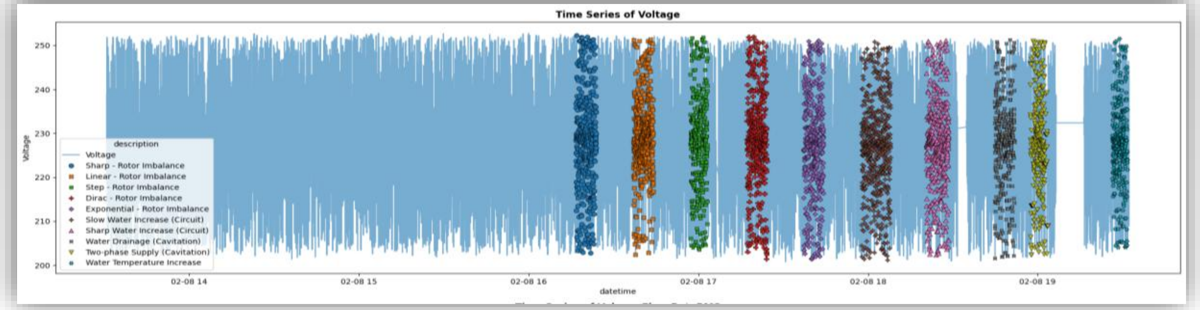


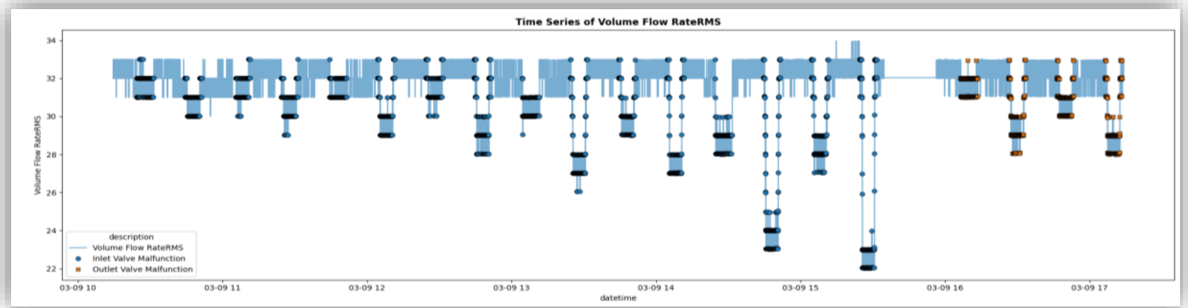
- 3D scatter plot shows unique cluster formations in comparison to normal data points and anomalies.
- For some features these clusters are overlapping – difficult to distinguish

Time-series Analysis









- Vibrational Features occasionally shows spikes during anomalies.
- Temperature Features do not show any significant variation for anomalies.
- Voltage, Current & Pressure do not show any significant variation for anomalies
- Rate of flow of Fluid occasionally shows downward spikes during anomalies.

Feature Engineering

Interaction Features

- **Electrical Resistance**

$$Electrical\ Resistance = \frac{Voltage}{Current}$$

- **Flow Resistance**

$$Flow\ Resistance = \frac{Pressure}{Rate\ of\ flow\ of\ Fluid}$$

- **Operational Temperature Ratio**

$$Operational\ Temperature\ Ratio = \frac{Temperature\ of\ Engine}{Temperature\ of\ Fluid}$$

- **Operational Vibration**

$$Operational\ Vibration = \frac{Vibration\ along\ x - axis}{Vibration\ along\ y - axis}$$

- **Specific Power**

$$Specific\ Power = \frac{Voltage\ * Current}{Rate\ of\ flow\ of\ Fluid}$$

Difference Features

- **Vibrational Balance**

$$Vibrational\ Balance = Vibration\ (along\ y - axis)\ - Vibration\ (along\ x - axis)$$

- **Temperature Balance**

$$Temperature\ Balance = Temperature\ of\ Engine\ - Temperature\ of\ Fluid$$

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Efficiency & Performance Features

- **Hydraulic Power**

$$Hydraulic\ Power = \frac{(Flow\ Rate\ * Fluid\ Density\ * g\ * Differencial\ Head)}{1000}$$

- **Electrical Power**

$$Electrical\ Power = Voltage\ * Current\ * Power\ Factor$$

- **Pump Efficiency**

$$Pump\ Efficiency = \frac{Hydraulic\ Power}{Electrical\ Power}$$

Higher Order Interactive Features

- **Vibration Intensity Ratio**

$$Vibration\ Intensity\ Ratio = \frac{Vibration\ (along\ x - axis)}{Vibration\ (along\ y - axis)}$$

- **Vibration per Unit Flow**

$$Vibration\ per\ Unit\ Flow = \frac{Operational\ Vibration}{Rate\ of\ flow\ of\ Fluid}$$

- **Engine Temperature per Unit Power**

$$Engine\ Temperature\ per\ Unit\ Power = \frac{Engine\ Temperature}{Power}$$

- **Efficiency & Temperature Balance Ratio**

$$Efficency\ \&\ Temperature\ Balance\ Ratio = \frac{Pump\ Efficiency}{Temperature\ Balance}$$

Modelling & Evaluation

Statistical Models

Technique	Name	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Parametric	Z-Score	**0.923**	0.766	**0.077**	0.234	0.726	0.548	0.234	0.328
	Multivariate Gaussian	0.830	0.376	0.170	0.625	0.771	**0.594**	0.624	0.609
Non-parametric	Inter Quartile Range	0.802	0.584	0.198	0.416	0.692	0.457	0.416	0.436
	Percentile	0.775	**0.246**	0.225	**0.754**	0.769	0.573	**0.754**	**0.651**
Proximity-Based	k-Nearest Neighbors	0.795	0.462	0.205	0.538	**0.772**	0.512	0.538	0.525
	Local Outlier Factor	0.789	0.291	0.211	0.709	0.766	0.573	0.709	0.634

- Z-Score Method
- Multivariate Gaussian Model
- Interquartile Range (IQR)
- Percentile Method
- k-Nearest Neighbors
- Local Outlier Factor

• The Percentile method stands out as the best-performing model by a significant margin, demonstrating superior accuracy, precision, recall, and F1-score due to its excellent balance of true positives and true negatives, and low false positive rates.

• The Local Outlier Factor (LOF) is the second-best model, showing good overall performance.

• The Z-Score method, while excellent at detecting true positives, suffers severely from high false positives, making it unreliable.

• The Inter Quartile Range (IQR) and k-Nearest Neighbors (k-NN) methods are also weak performers, showing poor results across most evaluation metrics.

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Machine Learning Models

Technique	Name	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Unsupervised	K-Means Clustering	0.206	0.085	0.794	0.915	0.408	0.315	0.915	0.469
	Isolation Forest	0.768	0.563	0.232	0.437	0.673	0.429	0.437	0.433
	One-Class SVM	**0.995**	0.972	**0.006**	0.028	0.719	0.670	0.028	0.054
Supervised	Logistic Regression	0.985	0.371	0.015	0.629	0.884	0.945	0.629	0.755
	Support Vector Machine	0.971	0.267	0.029	0.733	0.903	0.910	0.733	0.812
	Random Forest Classifier	0.992	0.111	0.009	0.889	0.962	**0.977**	0.889	0.931
	XGBoost	0.986	**0.082**	0.014	**0.918**	**0.967**	0.964	**0.918**	**0.940**

- K-Mean Clustering
- Isolation Forest
- One-Class SVM
- Logistic Regression
- SVM
- Random Forest Classifier
- XGBoost Classifier

• Supervised models significantly outperform unsupervised models in this comparison, especially in terms of overall accuracy and balanced F1-Scores.

• XGBoost is the best-performing model overall, exhibiting exceptional balance and high scores across all key metrics (Accuracy, Precision, Recall, F1-Score).

• Random Forest Classifier is a very close second to XGBoost, also demonstrating excellent and well-balanced performance.

• Support Vector Machine is a strong contender within supervised learning, offering robust performance.

Deep Learning Models

Models	Std. Threshold	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Autoencoder	0.5	0.9715	**0.9734**	0.0285	**0.0266**	0.702	0.272	**0.027**	**0.049**
	1	0.9941	0.9944	0.0059	0.0056	0.712	0.277	0.006	0.011
	2	0.9962	0.9962	0.0038	0.0038	**0.713**	0.287	0.004	0.008
	3	0.997	0.9969	0.003	0.0031	**0.713**	**0.297**	0.003	0.006
Variational Autoencoder	0.5	0.9783	0.9797	0.0217	0.0203	0.705	0.272	0.02	0.038
	1	0.9974	0.9974	0.0026	0.0026	**0.713**	0.285	0.003	0.005
	2	0.9976	0.9979	0.0024	0.0021	**0.713**	0.254	0.002	0.004
	3	**0.9977**	0.9984	**0.0023**	0.0016	**0.713**	0.219	0.002	0.003

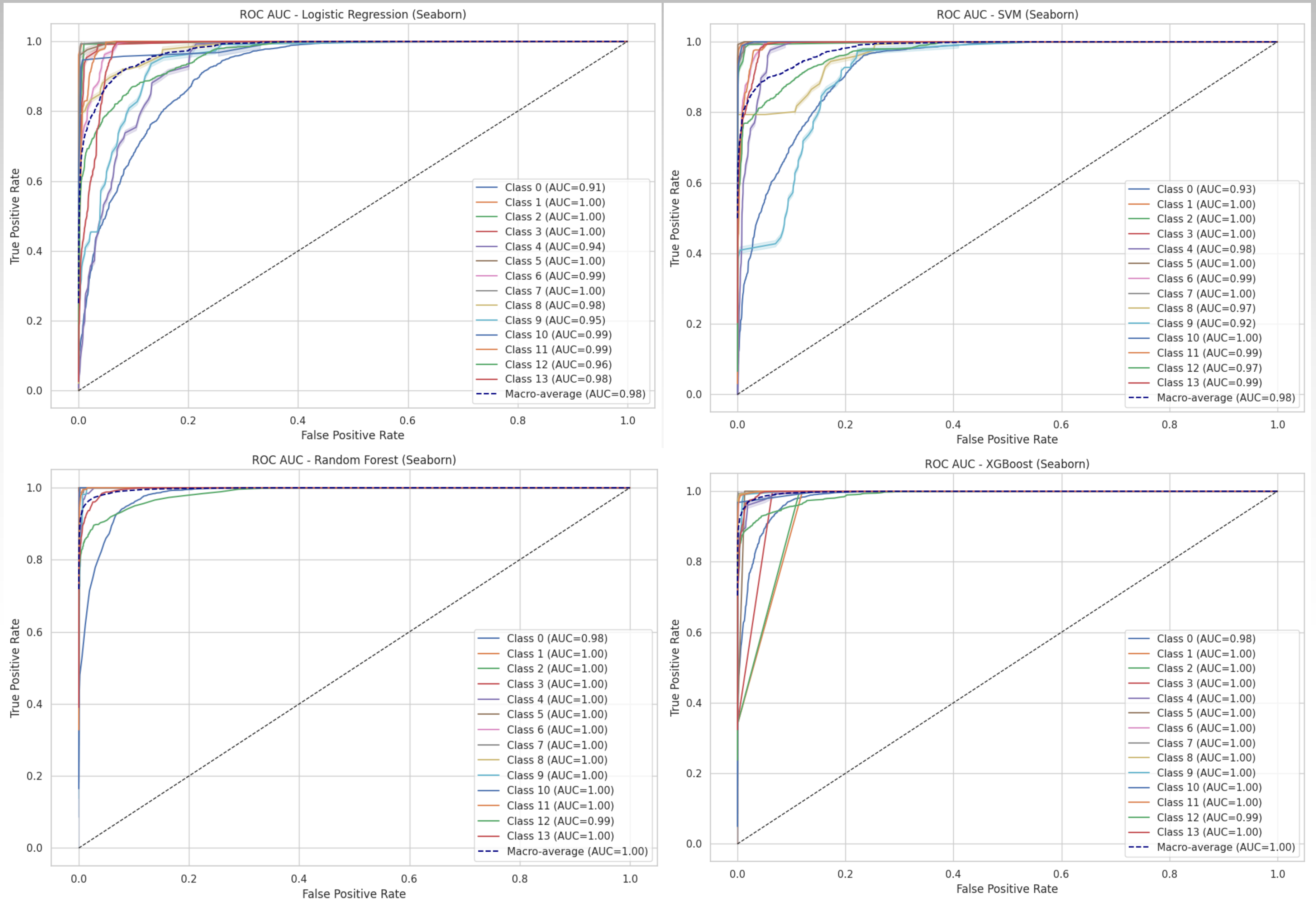
- Autoencoder
- Variational Autoencoder

• Both AE and VAE models show extremely high True Positive (TP) rates but are severely hampered by very high False Positive (FP) rates across all tested thresholds.

• Consistently low Precision and extremely low F1-Scores for both models, indicates a poor balance between detecting true anomalies and minimizing false alarms.

• The Recall values are also extremely low, which, when combined with high True Positive (TP), strongly suggests an underlying issue in detecting such anomalies.

• Contrary to expectation, VAE did not perform better than AE; the latter achieved a slightly higher F1-Score and marginally better precision.



Additional Evaluation Plots (GridSearchCV)

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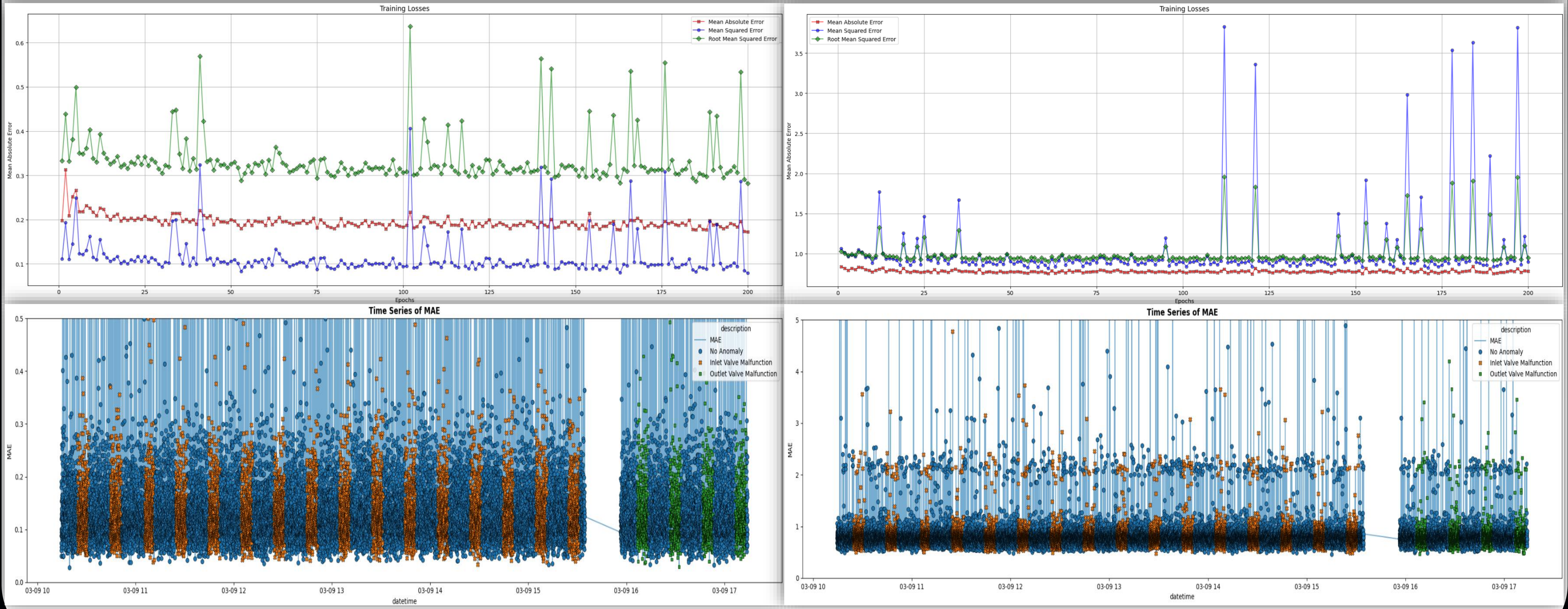
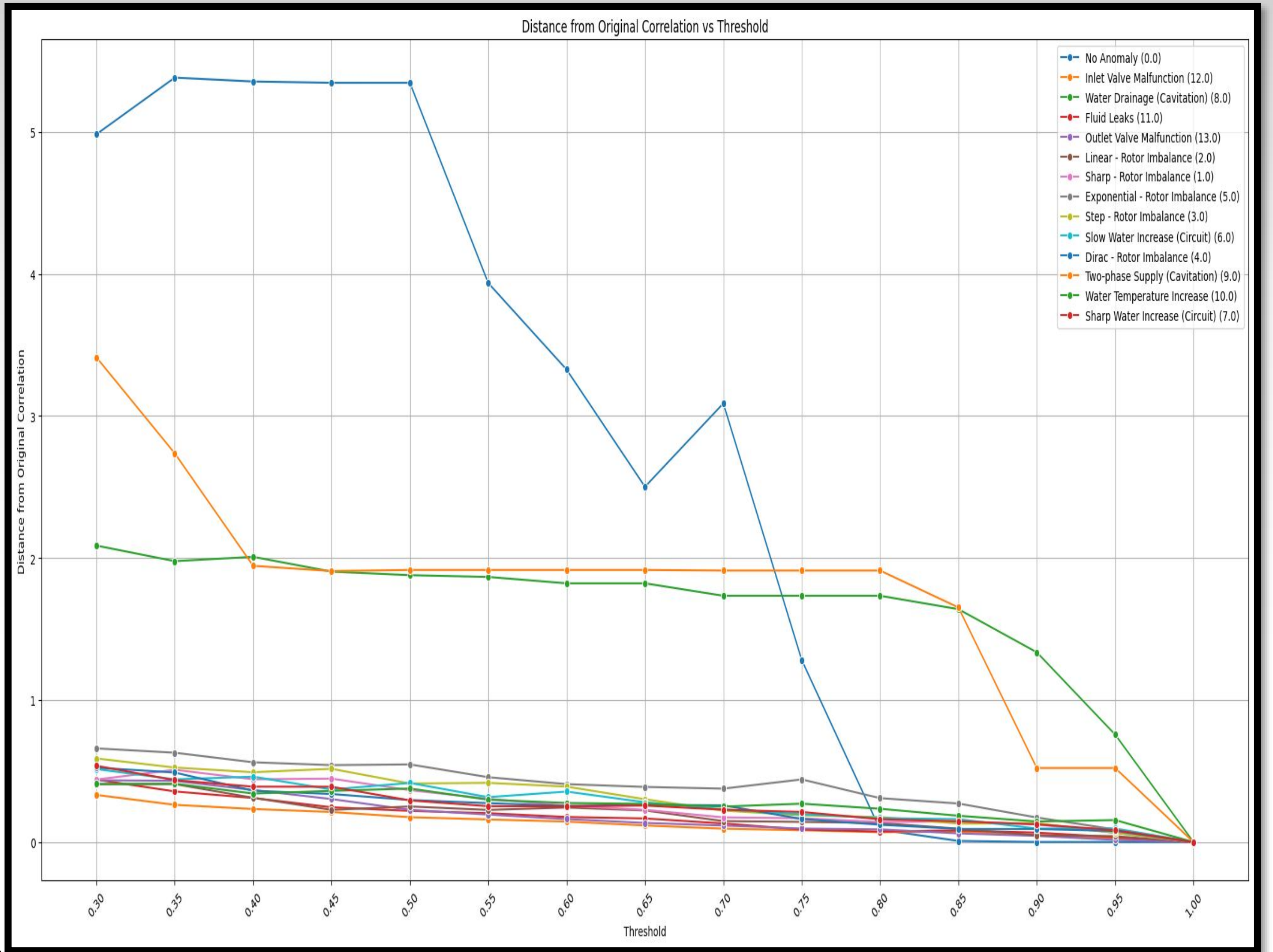
Additional Evaluation Plots



Synthetic Data Generation

Under-sampling



- Undersampling is used to remove outliers within data clusters before synthetic data generation, preventing outliers from distorting class-specific distributions.
- Accurate prediction of minority classes, especially in overlapping clusters like in the SKAB dataset, is critical; undersampling improves minority class generalization but may reduce overall accuracy due to information loss.
- The SKAB dataset's 14 classes (1 normal, 13 anomalies) exhibit overlapping clusters caused by varying operational states of the centrifugal pump, necessitating a tailored undersampling approach.
- Undersampling involves computing cluster centers, measuring and normalizing Euclidean distances of points to centers, then applying a threshold to filter outliers and overlapping points without disturbing cluster integrity.
- A threshold of 0.8 on normalized distances best preserves sensor parameter correlations across clusters, balancing outlier removal and data fidelity for effective undersampling.

Sampling-Based Techniques

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- K-Mean Clustering
- Isolation Forest
- One-Class SVM
- Logistic Regression
- SVM
- Random Forest Classifier
- XGBoost Classifier

SMOTE

Technique	Name	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Unsupervised	K-Means Clustering	0.174	**0.055**	0.826	0.945	0.394	0.314	0.945	0.471
	Isolation Forest	0.520	0.367	0.480	0.633	0.552	0.345	0.633	0.446
	One-Class SVM	**0.961**	0.943	**0.039**	0.057	0.703	0.368	0.057	0.099
Supervised	Logistic Regression	0.412	0.041	0.588	**0.959**	0.568	0.395	0.959	0.559
	Support Vector Machine	-	-	-	-	-	-	-	-
	Random Forest Classifier	0.956	0.082	0.044	0.918	**0.945**	**0.894**	0.918	**0.906**
	XGBoost	0.930	0.063	0.071	0.932	0.930	0.841	**0.932**	0.884

ADASYN

Technique	Name	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Unsupervised	K-Means Clustering	0.120	0.046	0.881	0.954	0.358	0.302	0.954	0.459
	Isolation Forest	0.552	0.353	0.448	0.647	0.579	0.366	0.647	0.467
	One-Class SVM	**0.954**	0.911	**0.046**	0.089	0.707	0.437	0.089	0.148
Supervised	Logistic Regression	0.300	**0.008**	0.700	**0.992**	0.497	0.361	0.992	0.530
	Support Vector Machine	-	-	-	-	-	-	-	-
	Random Forest Classifier	0.952	0.076	0.048	0.924	**0.944**	**0.885**	0.924	**0.904**
	XGBoost	0.922	0.064	0.078	0.936	0.926	0.828	**0.936**	0.879

- Both SMOTE and ADASYN improve minority class detection and overall model balance versus baseline results.
- Random Forest and XGBoost consistently perform best under both oversampling methods, showing high reliability and robustness.
- Ensemble supervised models maintain strong precision, recall, and F1-scores across sampling strategies.
- SMOTE yields slightly better overall results for Random Forest, especially in balanced metric performance.
- ADASYN offers marginal improvements in recall for unsupervised models but struggles with precision and F1-score.
- Unsupervised approaches remain less effective compared to supervised ensembles, even after oversampling

Generative Modeling Technique

- K-Mean Clustering
- Isolation Forest
- One-Class SVM
- Logistic Regression
- SVM
- Random Forest Classifier
- XGBoost Classifier

Conditional Wasserstein GAN - GP

Technique	Name	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Unsupervised	K-Means Clustering	0.000	0.089	1.000	0.911	0.260	0.267	0.911	0.413
	Isolation Forest	0.620	0.487	0.380	0.513	0.590	0.350	0.513	0.416
	One-Class SVM	0.964	0.942	**0.036**	0.058	0.706	0.394	0.058	0.101
Supervised	Logistic Regression	0.279	**0.088**	0.721	**0.912**	0.460	0.336	**0.912**	0.491
	Support Vector Machine	-	-	-	-	-	-	-	-
	Random Forest Classifier	**0.957**	0.104	0.043	0.896	**0.940**	**0.893**	0.896	**0.894**
	XGBoost	0.952	0.101	0.048	0.900	0.937	0.882	0.899	0.891

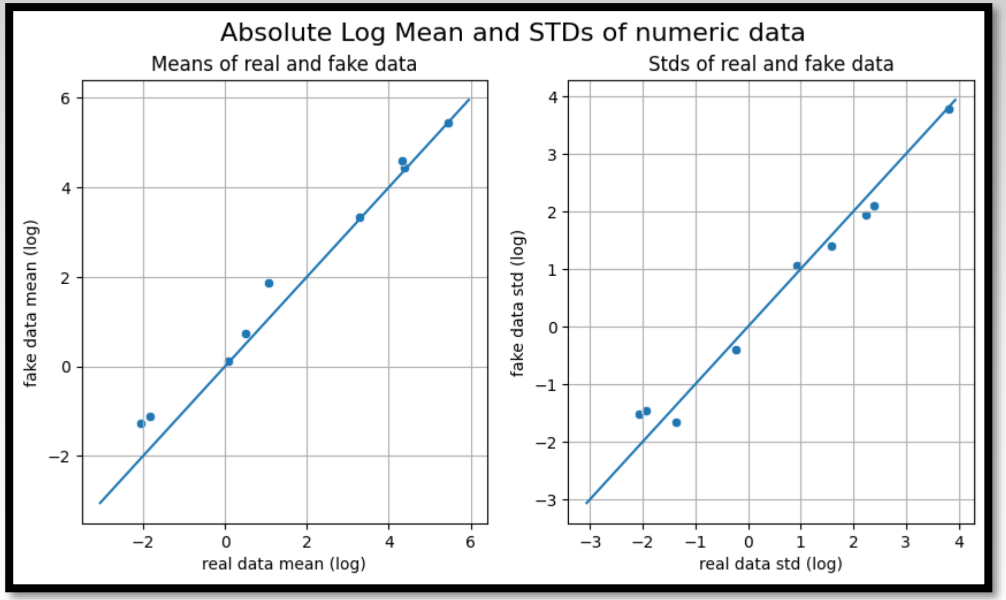
Spectral Normalized Conditional Wasserstein GAN - GP

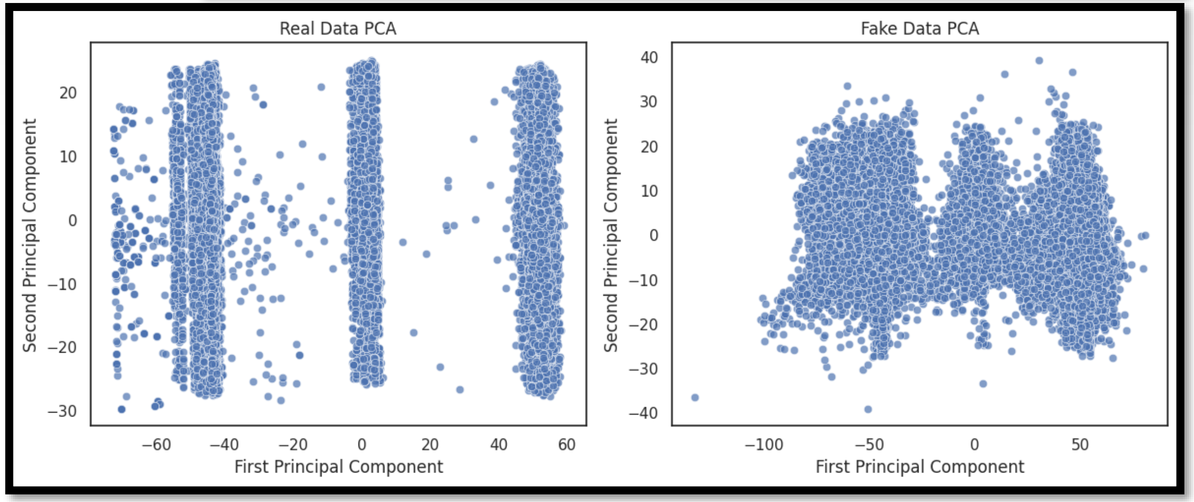
Technique	Name	TP	FP	FN	TN	Accuracy	Precision	Recall	F1-Score
Unsupervised	K-Means Clustering	0.053	**0.033**	0.947	**0.967**	0.314	0.290	**0.967**	0.446
	Isolation Forest	0.165	0.044	0.835	0.956	0.391	0.314	0.956	0.472
	One-Class SVM	0.896	0.822	0.105	0.178	0.691	0.405	0.178	0.247
Supervised	Logistic Regression	0.934	0.344	0.066	0.656	0.854	0.798	0.656	0.720
	Support Vector Machine	-	-	-	-	-	-	-	-
	Random Forest Classifier	**0.958**	0.103	**0.042**	0.898	**0.941**	**0.896**	0.897	**0.897**
	XGBoost	0.948	0.106	0.052	0.894	0.933	0.873	0.894	0.884

- Unsupervised models generally demonstrate low Precision and F1-Scores, with high False Positives, limiting their practical efficacy post-synthetic balancing.
- Logistic Regression benefits notably from SN-CWGAN-GP, with marked F1-Score improvement over CWGAN-GP, reflecting enhanced synthetic data quality.
- Random Forest consistently achieves the highest Accuracy, Precision, Recall, and F1-Score across both CWGAN-GP and SN-CWGAN-GP methods.
- XGBoost performs robustly but marginally below Random Forest in key metrics under both GAN-based augmentations.
- SN-CWGAN-GP's spectral normalization stabilizes GAN training, leading to superior synthetic sample fidelity and better model generalization.
- Random Forest's ensemble nature secures robustness to imbalance and synthetic data variations, making it the optimal choice for balanced anomaly detection.

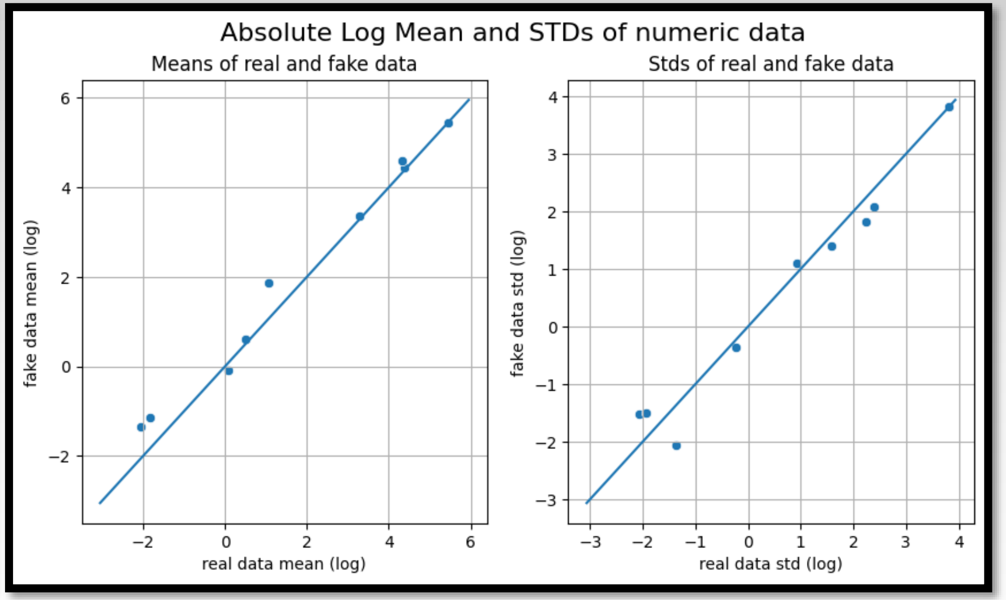
Data Fidelity

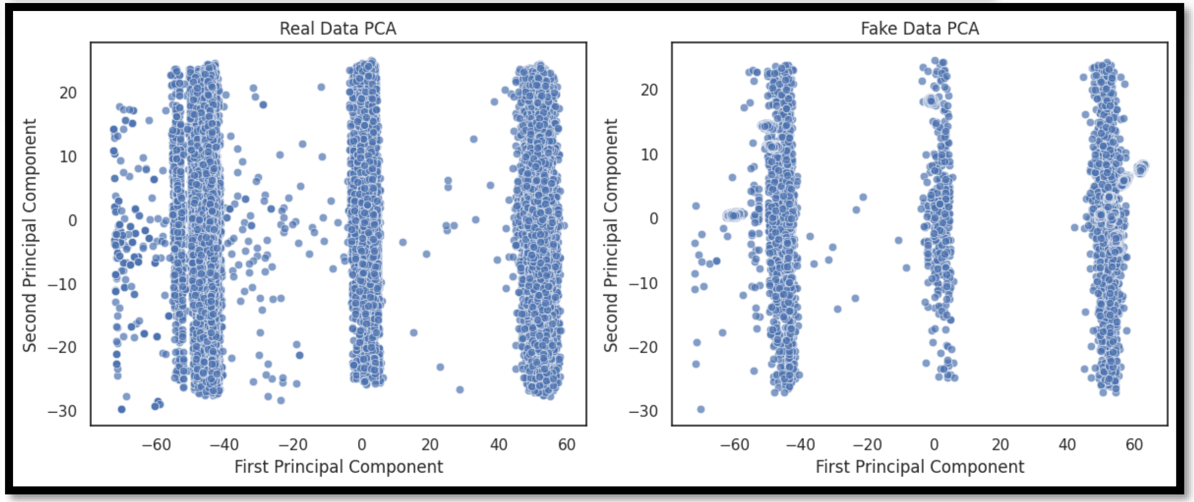
Conditional Wasserstein GAN - GP





Spectral Normalized Conditional Wasserstein GAN - GP





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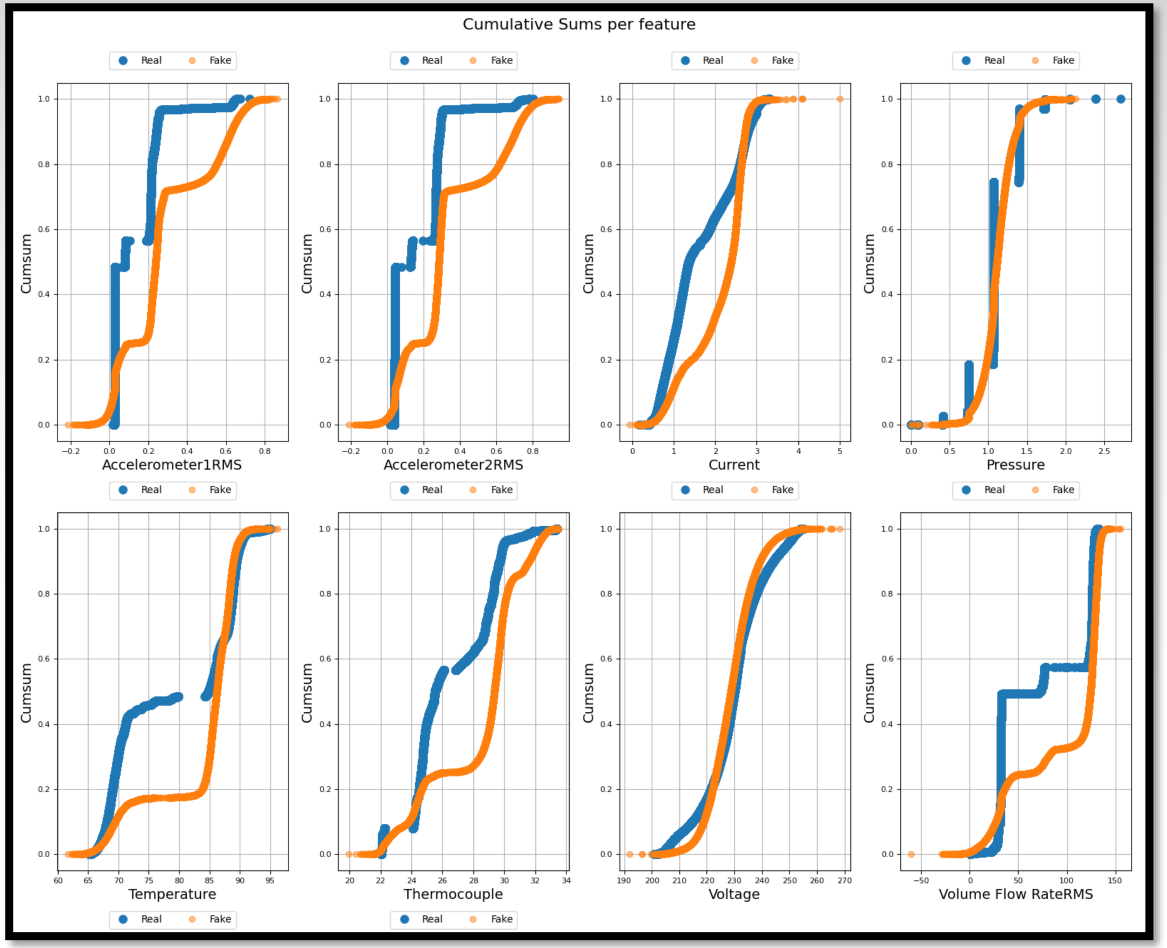
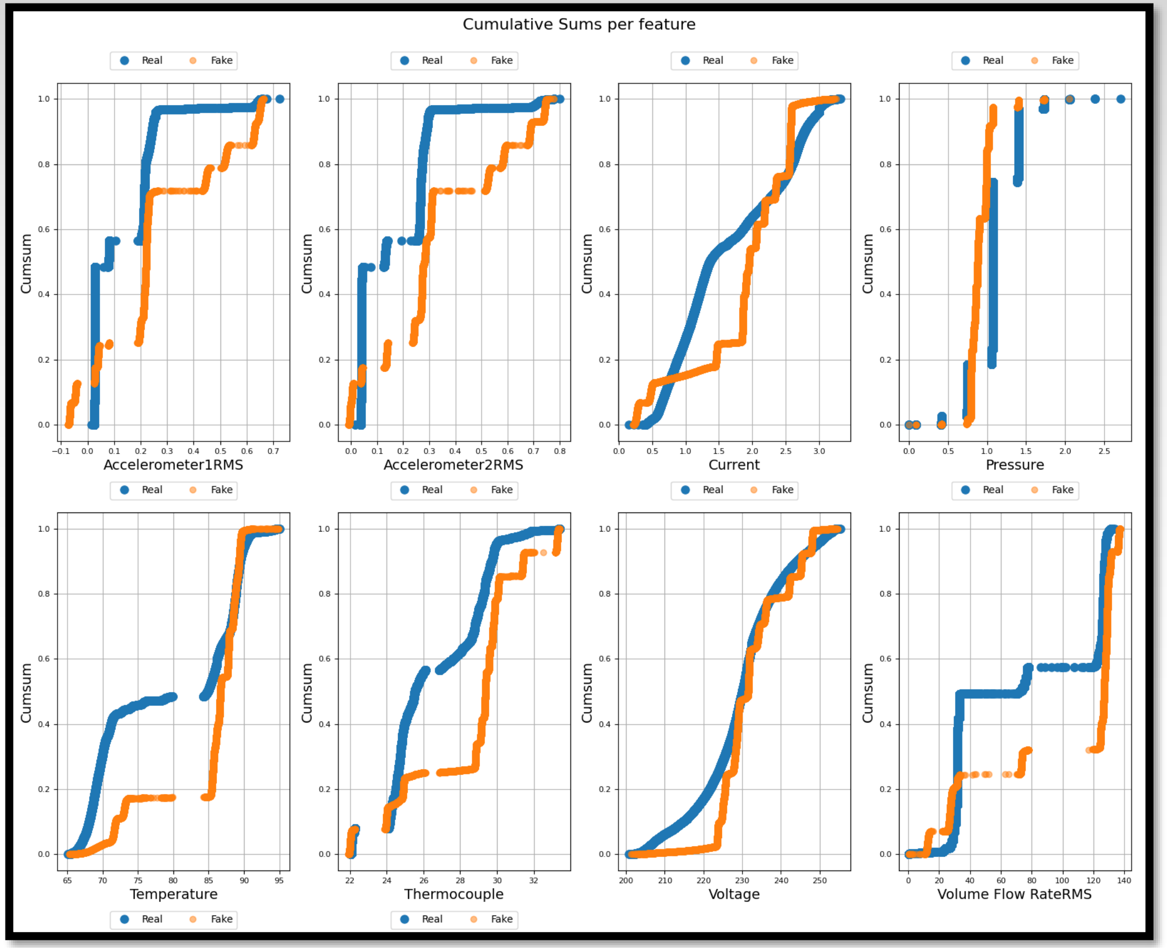
Conditional Wasserstein GAN - GP

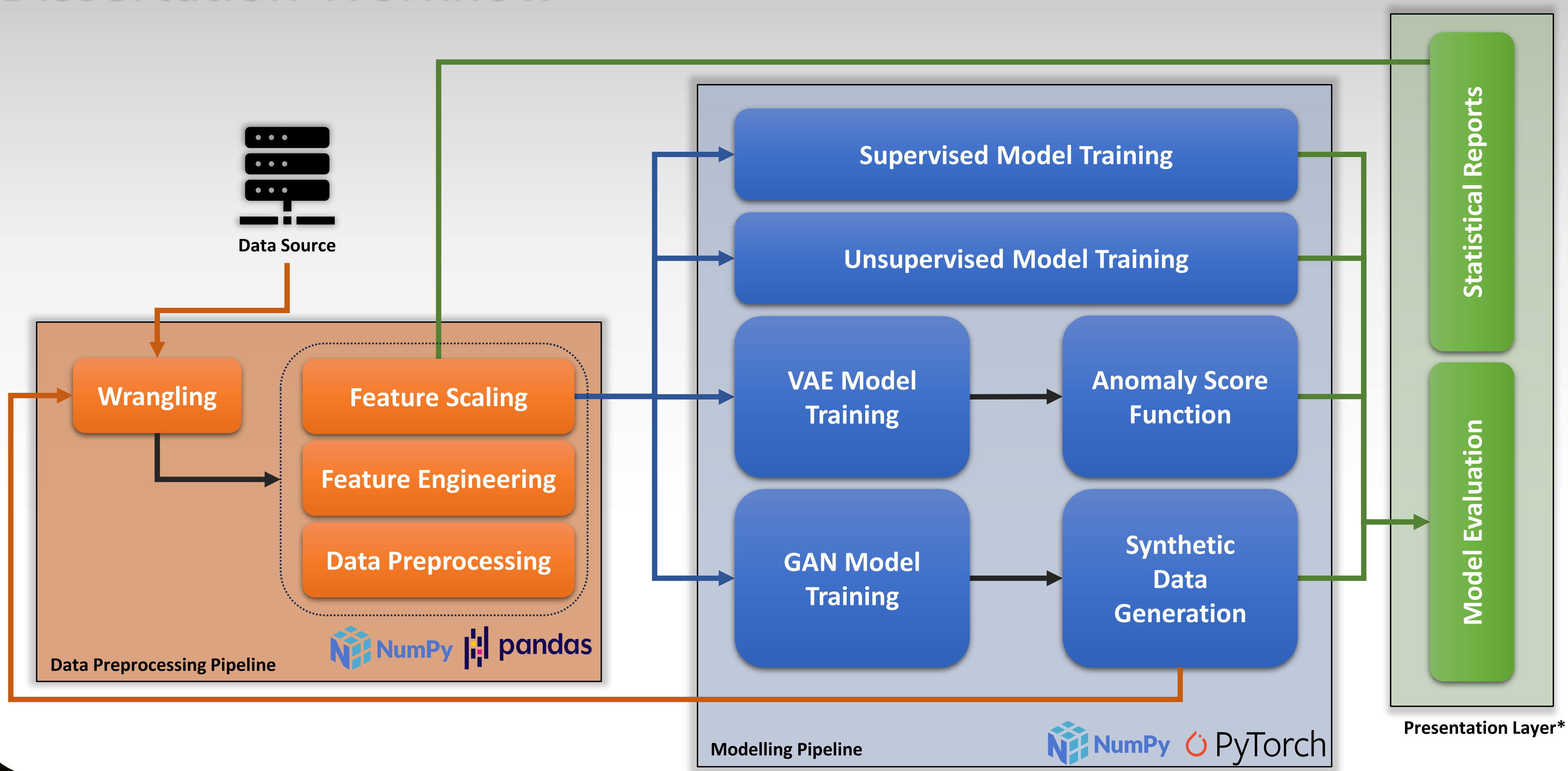


Spectral Normalized Conditional Wasserstein GAN - GP



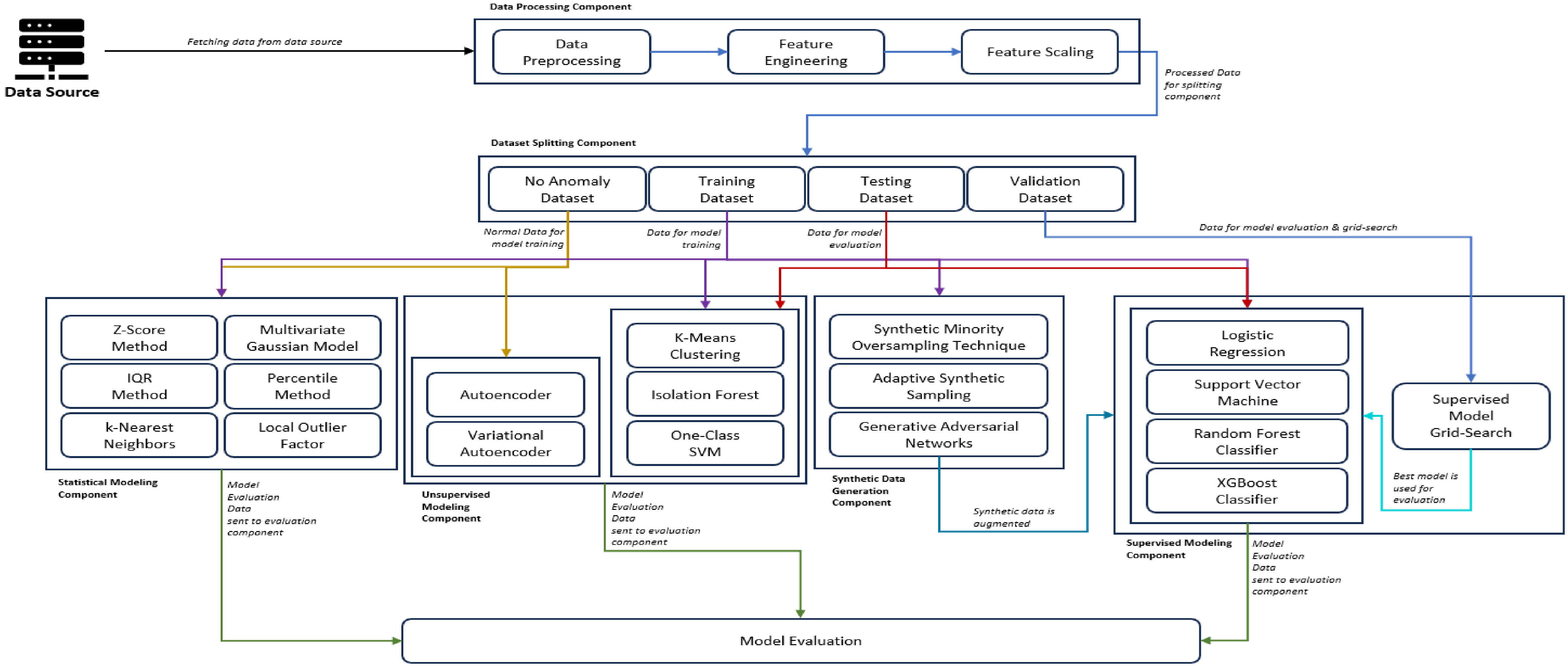
Solution Plan

Dissertation Workflow



*The technology to be used to represent the reports in the presentation layer are yet to be decided.

Data Flow Diagram



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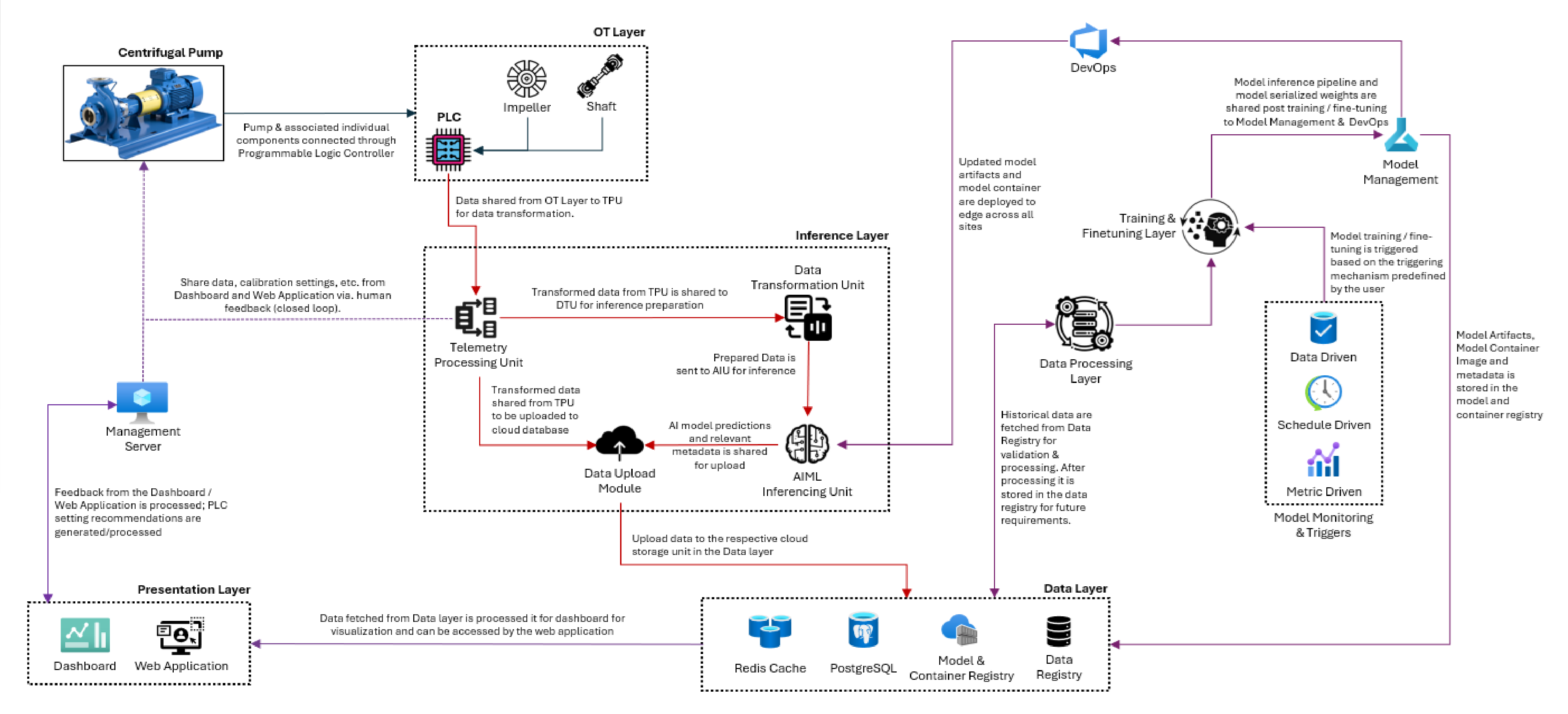
Proposed Production System Architecture



* The detailed architecture is considering a production system; it is indicative and is subject to changed based on the procured infra.

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Planner

Sl No.	Deliverables	Status	Start Week	End Week
1	Proposal Approval	Completed	1	1
2	Literature Review	Completed	1	1
3	Dataset Finalization	Completed	2	2
4	Data Wrangling	Completed	2	3
5	Data Sampling Pre-processing	Completed	3	3
6	Exploratory Data Analysis	Completed	3	3
7	Feature Engineering	Completed	4	5
8	Statistical Modeling	Completed	5	5
9	Statistical Model Evaluation	Completed	5	5
10	Machine Learning Modeling	Completed	6	6
11	Machine Learning Model Evaluation	Completed	6	6
12	Deep Learning Modeling	Completed	7	7
13	Deep Learning Model Evaluation	Completed	8	8
14	Generative Modeling	Completed	9	9
15	Generative Model Evaluation	Completed	10	10
16	Synthetic Data Generation	Completed	11	11
17	Synthetic Data Evaluation	Completed	12	12
18	Machine Learning Modeling with Synthetic Data	Completed	13	13
19	Machine Learning Model Evaluation	Completed	14	14
20	Comprehensive Evaluation Report	Completed	15	15
21	Dissertation Finalization	Completed	15	15
22	Submission	Completed	16	16

Conclusion

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innovate achieve lead

• The work tackles data scarcity in industrial human-machine collaboration using generative AI and robust anomaly detection to boost AI reliability and safety.

• Random Forest with SMOTE outperforms shallow and non-parametric methods under severe class imbalance, effectively detecting rare anomalies.

• Autoencoders and VAEs offer limited improvement, while class-conditional GANs match oversampling in synthesizing realistic anomalies for complex data.

• Combining generative models with feature engineering and synthetic augmentation enhances anomaly detection, operator training, and AI system validation in Industry 4.0.

• Advances in GAN architectures and multimodal data will improve synthetic sample fidelity, bridging real and simulated scenarios, and strengthening ML performance in critical industrial applications.

Thank You

QA & Demo